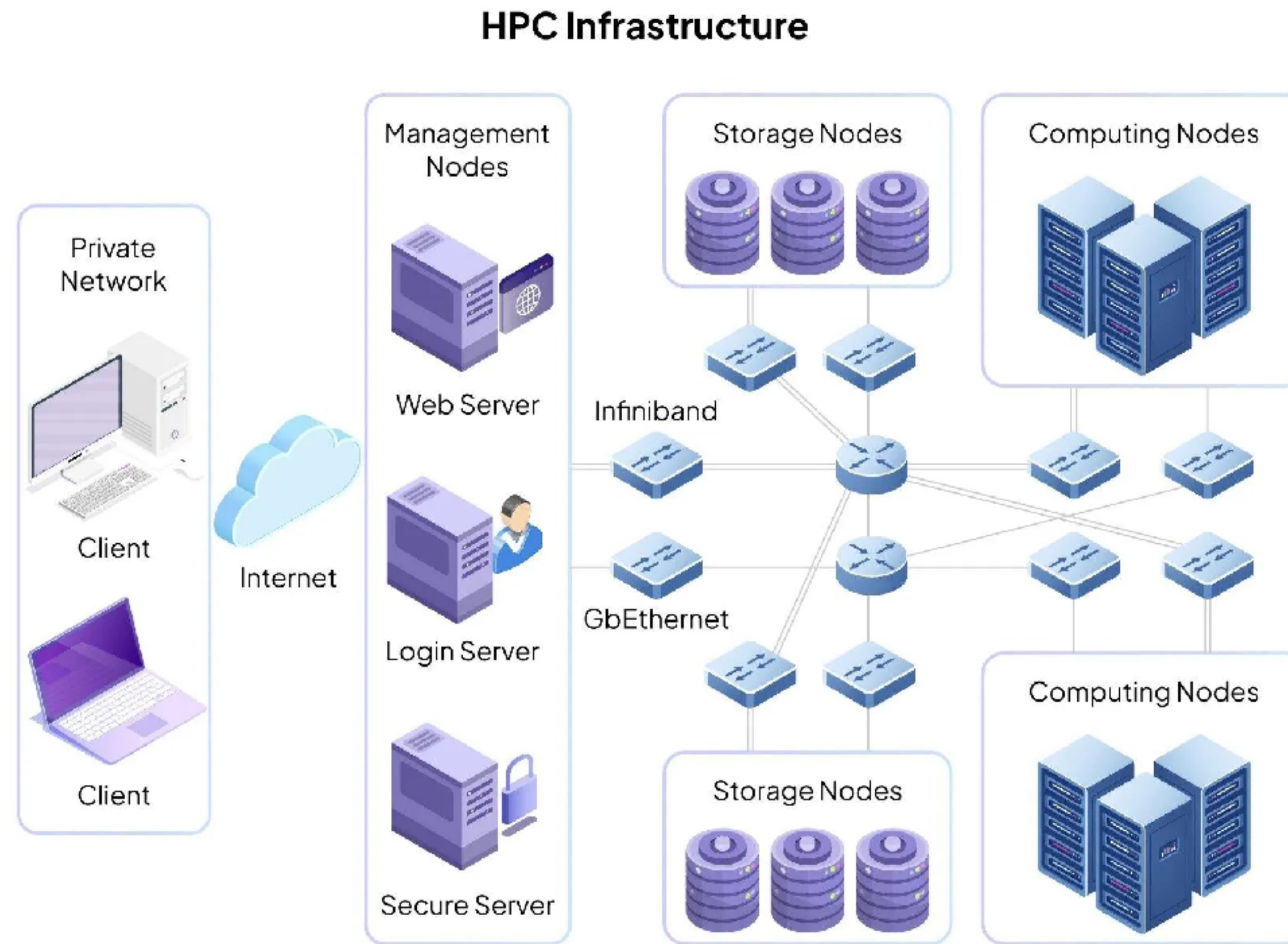


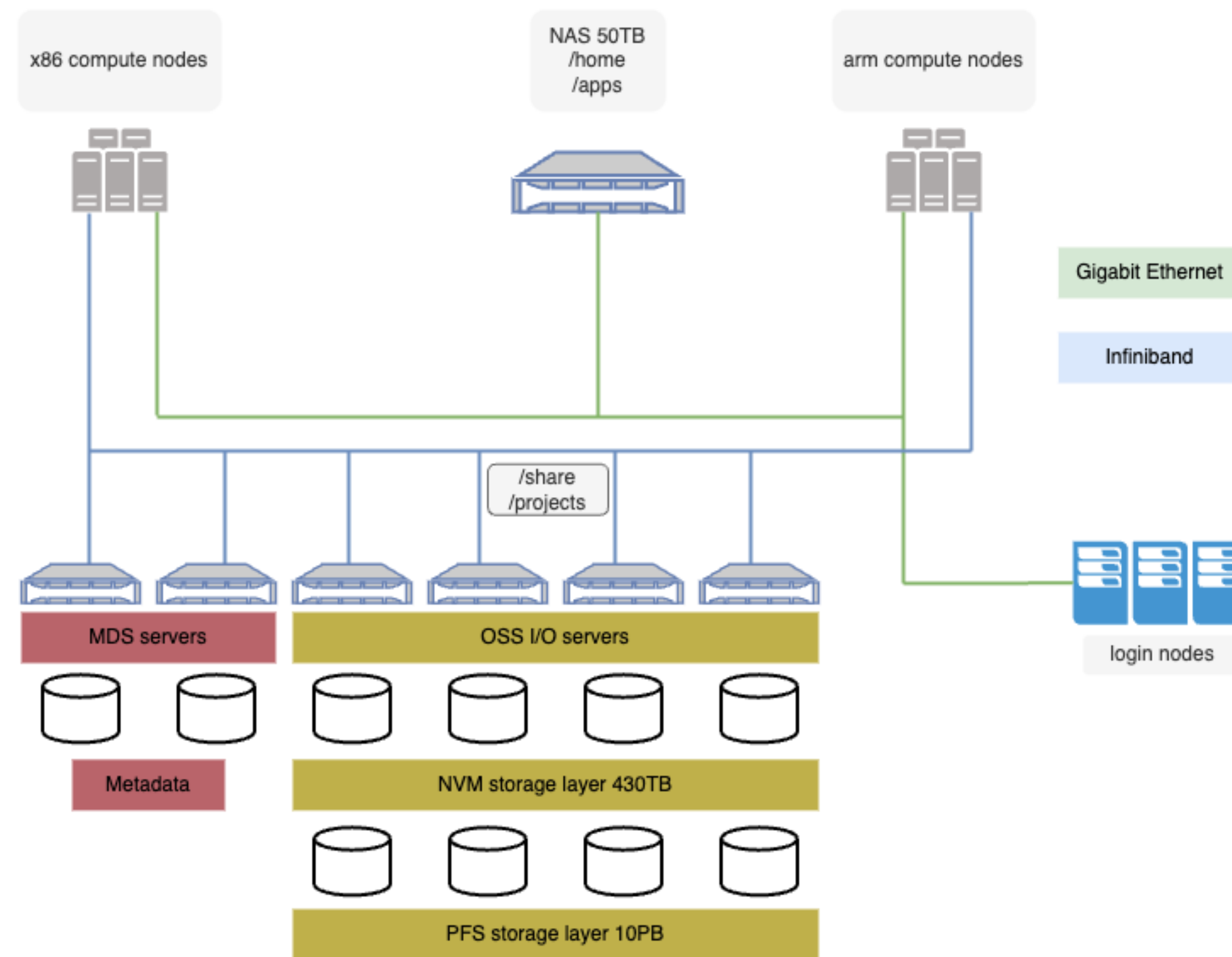
Arquiteturas de Sistemas

HPC

HPC Architecture



Arquitetura Deucalion



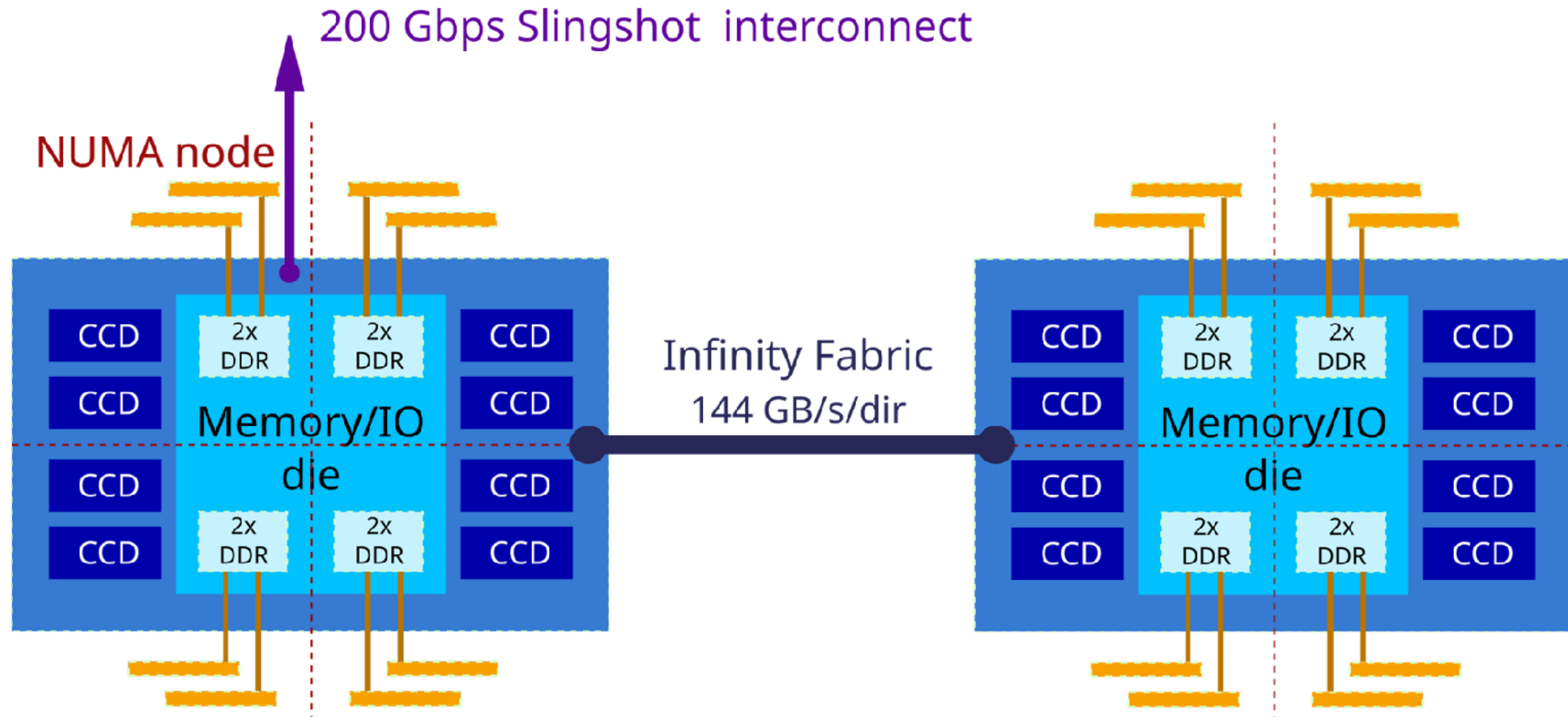
LUMI spec sheet: A modular system

L U M I

- LUMI-G: 2978 nodes with 1 AMD EPYC 7A53 CPU and 4 AMD MI250X accelerators (512 GB + 4x128 GB RAM)
- LUMI-C: 2048 nodes with 2 64-core AMD EPYC 7763 CPUs (1888x 256GB, 128x 512 GB and 32x 1TB)
- Nodes for interactive data analytics: 8 4TB CPU nodes and 8 nodes with 8 GPUs each for visualisation
- LUMI-F: 8 PB Lustre flash-based file storage (>2 TB/s)
- LUMI-P: 4 20 PB hard disk based Lustre file systems (4x 240 GB/s)
- 30 PB object based file system
- 4 user access nodes with two AMD Rome CPUs each for ssh access and some for web access via Open OnDemand
- All linked together with a HPE Cray Slingshot 11 interconnect

LUMI-C node

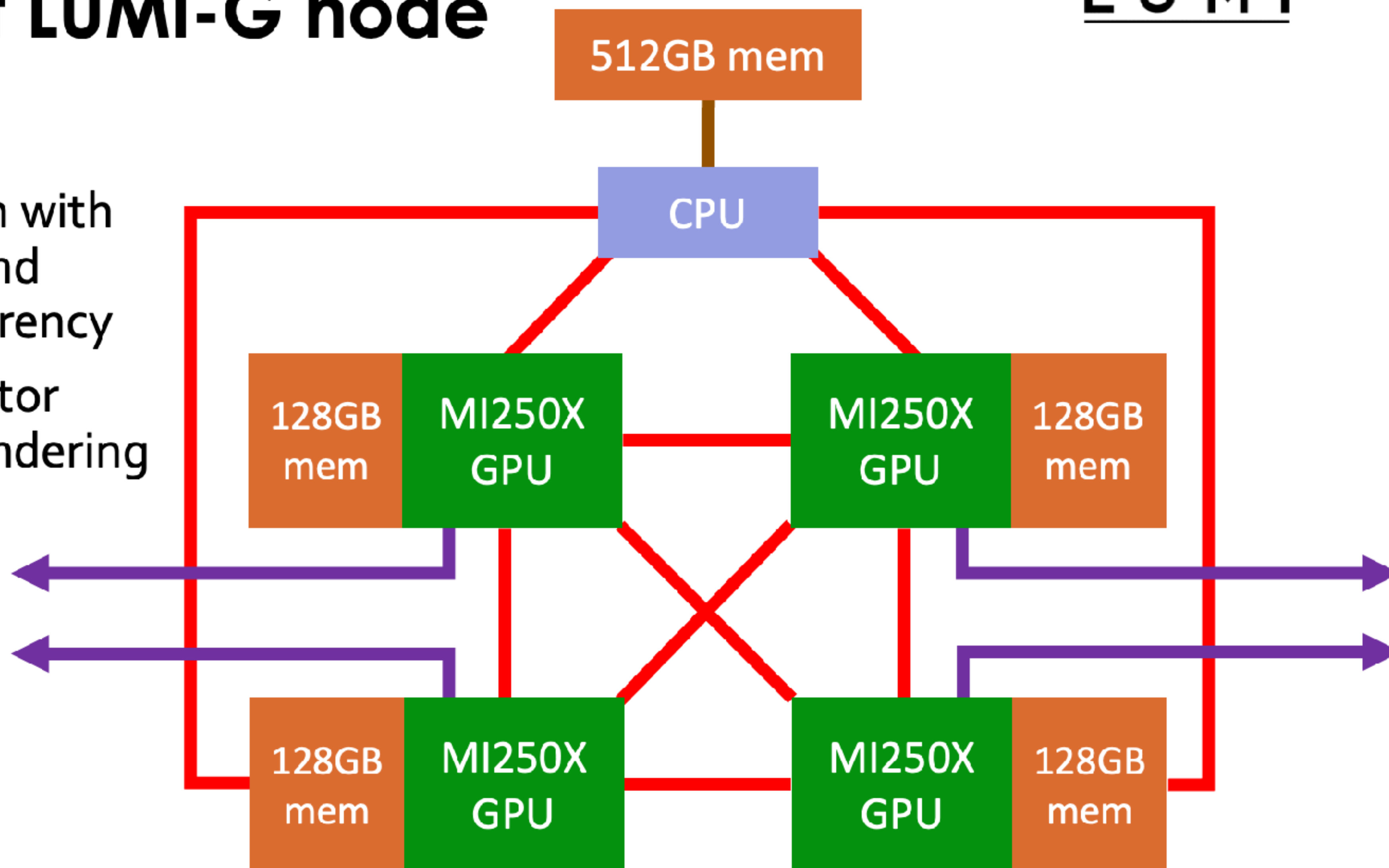
LUMI

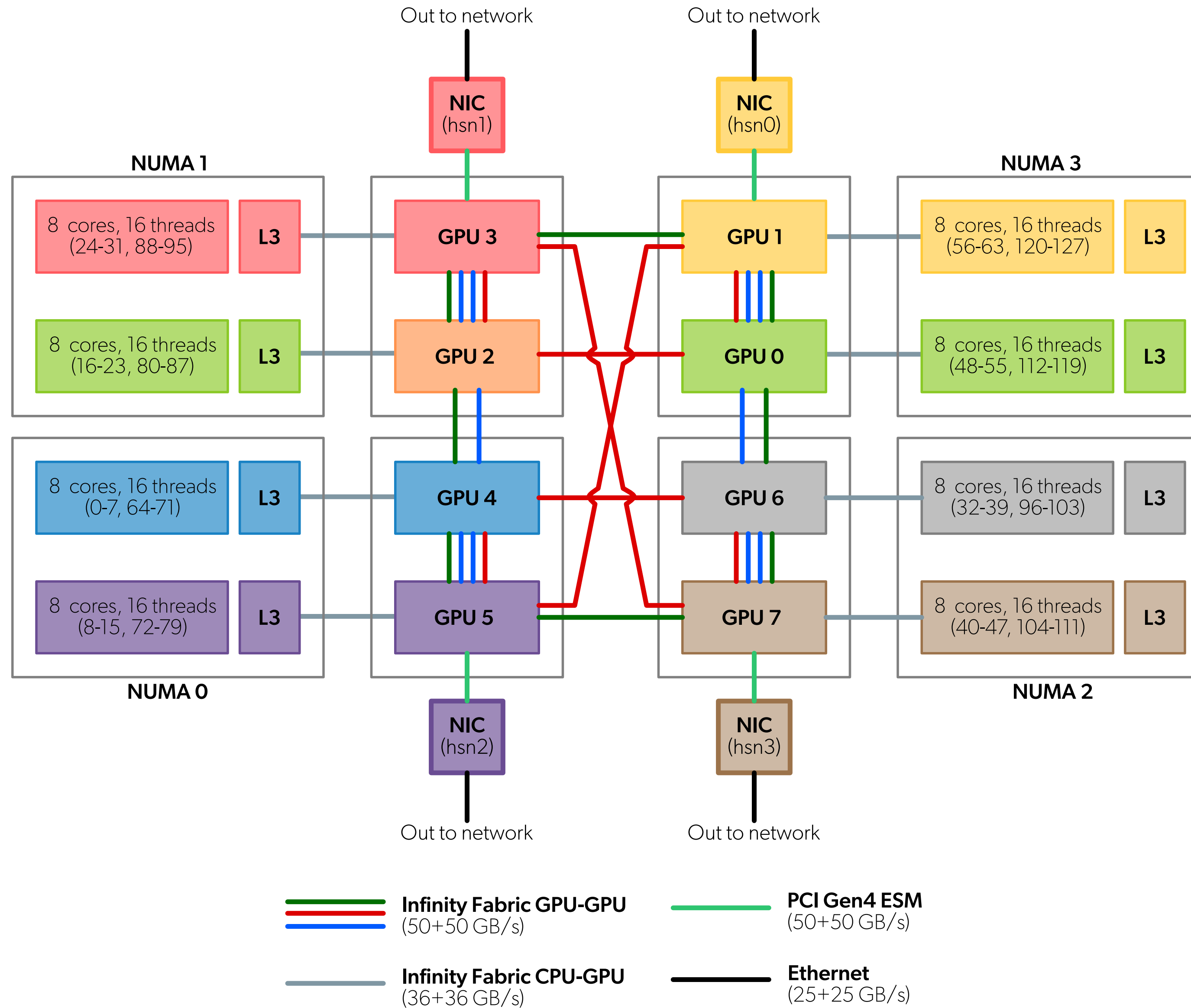


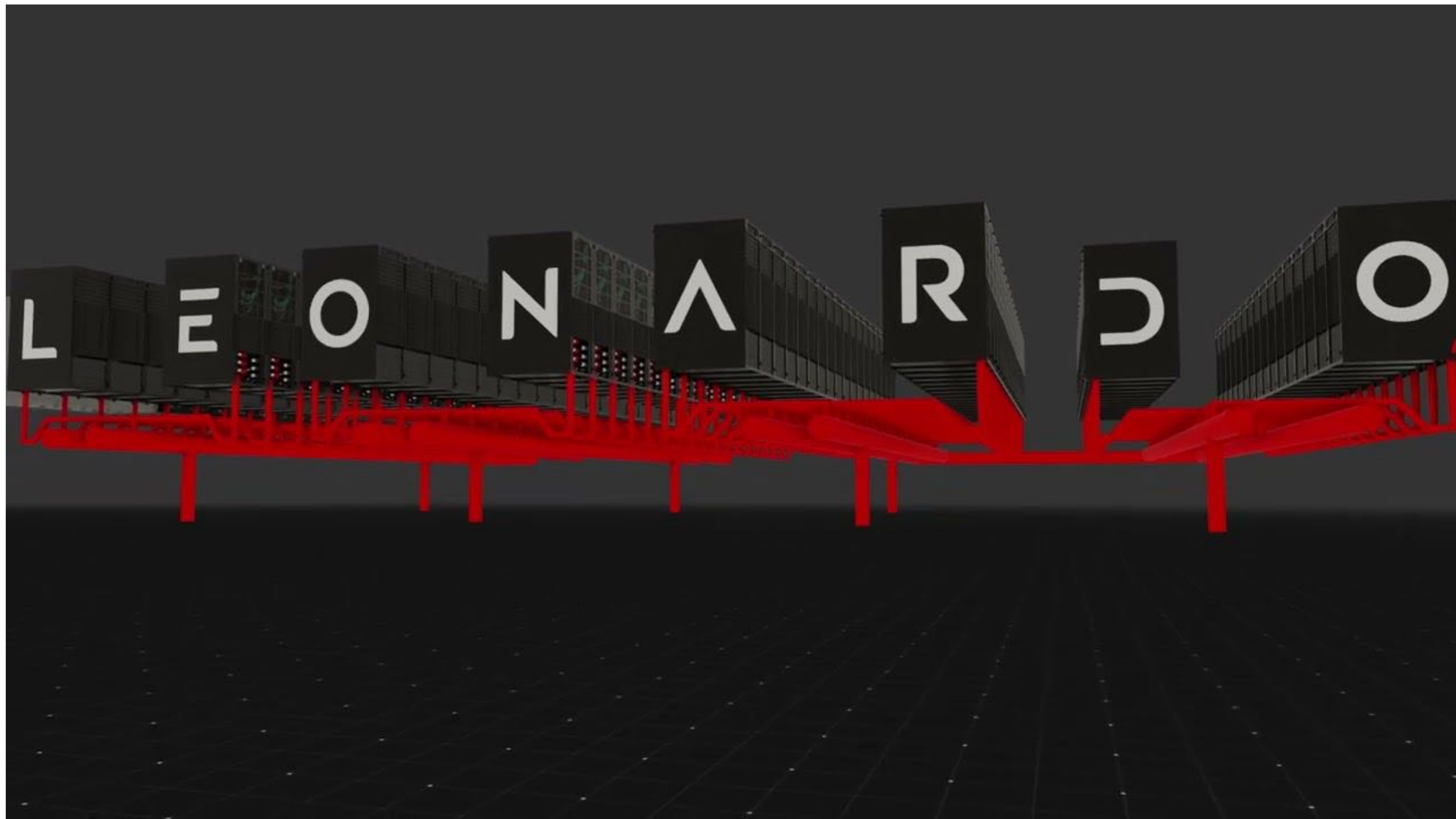
Concept LUMI-G node

LUMI

- “GPU first” system with unified memory and partial cache coherency
- Compute accelerator (CDNA2), not a rendering GPU (RDNA architecture)!







Leonardo Architecture

